House Price Analysis and Feature Exploration

1. Introduction

House price analysis helps us understand the factors associated with residential property prices. This project uses data analysis and visualization to explore house prices and their relationship with different property features.

2. Project Objectives

- To explore the house price dataset.
- To identify and handle missing values and duplicate records.
- To create visualizations to understand house price patterns.
- To identify important features associated with house prices.
- To provide useful business insights and recommendations.

3. Dataset Description

The dataset is from the Kaggle House Prices: Advanced Regression Techniques competition. The target variable for this project is SalePrice.

4. Data Cleaning and Preprocessing

- Checked the dataset structure and data types.
- Identified missing values.
- Filled missing numerical values using the median.
- Filled missing categorical values with the label "Missing".
- Checked and removed duplicate rows.
- Saved the cleaned dataset as "house_price_cleaned.csv".

5. Exploratory Data Analysis

Ten visualizations were created to explore:

1.

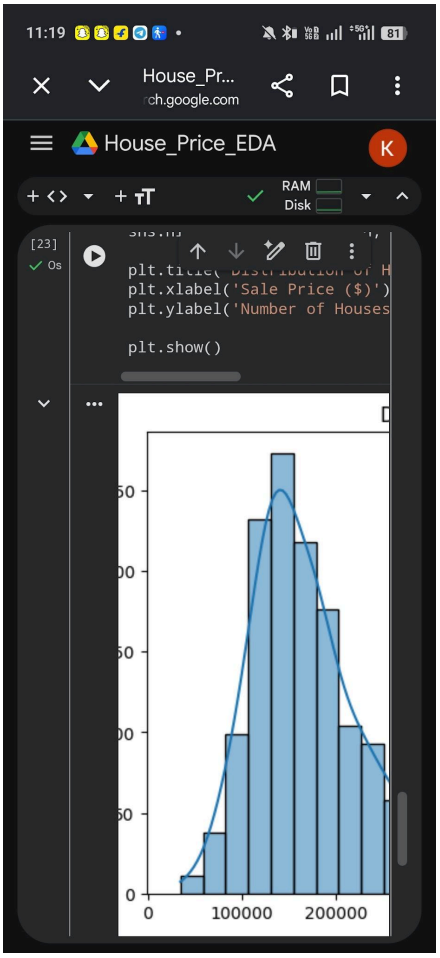


Figure 1: Distribution of House Prices

The histogram shows the distribution of house sale prices and helps identify the common price range and the presence of unusually expensive houses.

2.

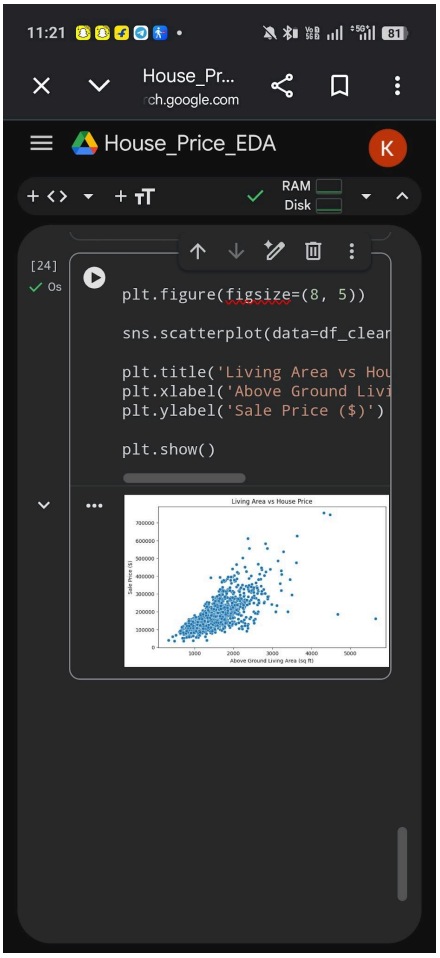


Figure 2: Living Area vs House Price

The scatter plot shows the relationship between above-ground living area and house sale price. The correlation analysis indicates a strong positive relationship between these two variables.

3.

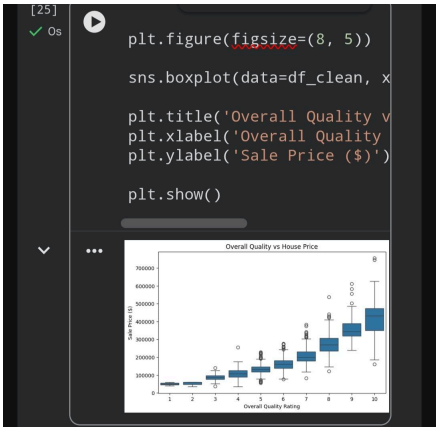


Figure 3: Overall Quality vs House Price

The box plot compares house sale prices across different overall quality ratings. The correlation value of 0.791 indicates a strong positive relationship between overall quality and sale price.

4. 

Figure 4: Number of Bedrooms vs House Price

This box plot compares house sale prices across different numbers of bedrooms. It helps explore how bedroom count is associated with house prices.

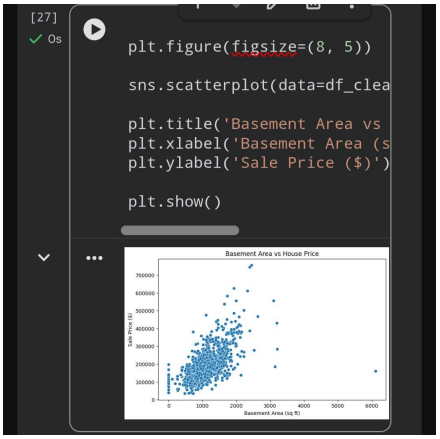
5. 

Figure 5: Basement Area vs House Price

This scatter plot explores the relationship between basement area and house sale price. The correlation coefficient of 0.614 indicates a positive relationship between these features.

6. 

Figure 6: Garage Capacity vs House Price

This box plot compares house sale prices based on garage capacity. The correlation coefficient of 0.640 indicates a positive relationship between garage capacity and sale price.

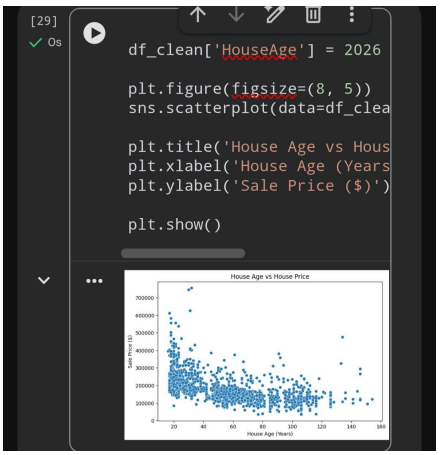
7. 

Figure 7: House Age vs House Price

This scatter plot explores the relationship between house age and sale price. It helps identify whether older and newer houses tend to have different price patterns.

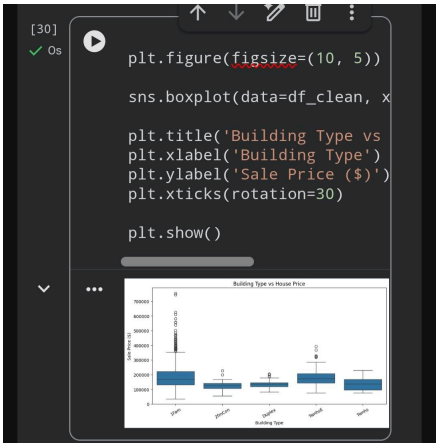
8. 

Figure 8: Building Type vs House Price

This box plot compares house sale prices across different building types. It helps explore how property type is associated with sale price.

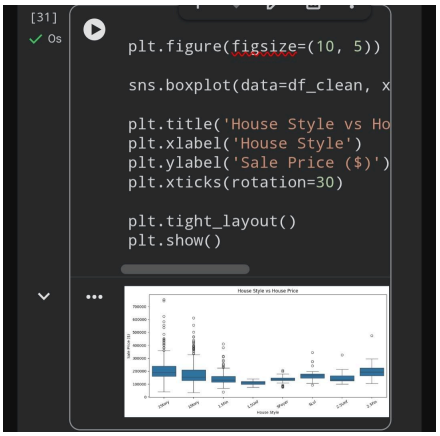
9. 

Figure 9: House Style vs House Price

This box plot compares house sale prices across different house styles. It helps identify differences in price distributions among property styles.

10. 

Figure 10: Correlation Heatmap of House Features

The correlation heatmap shows the relationships between numerical variables in the dataset. OverallQual (0.791) and GrLivArea (0.709) have strong positive correlations with SalePrice, indicating that these features are useful for further house price analysis.

6. Observations

1. Average House Price: The average house sale price is $180,921.20.

2. Median House Price: The median sale price is $163,000, which is lower than the average. This suggests that some expensive houses may be increasing the average price.

3. Overall Quality: Overall quality has a strong positive correlation with sale price (0.791).

4. Living Area: Above-ground living area has a strong positive correlation with sale price (0.709).

5. Other Features: Garage capacity (0.640), basement area (0.614), and year built (0.523) have positive correlations with sale price.

7. Business Insights

1. Importance of House Quality: Houses with higher overall quality tend to have higher sale prices. Real estate sellers can highlight construction quality and property condition.

2. Value of Living Space: Houses with larger living areas tend to have higher prices. Living area is a useful feature when comparing properties.

3. Importance of Additional Features: Garage capacity and basement area are positively associated with sale price. These features can help buyers compare properties and sellers position their houses in the market.

8. Recommendations

1. For Sellers: Maintain good house condition and highlight overall quality, living area, garage, and basement features in property listings.

2. For Buyers: Compare house quality, living area, garage capacity, and basement area before making a purchase decision.

3. For Real Estate Businesses: Develop a machine learning model using important house features to estimate sale prices. Evaluate the model's accuracy before using predictions for real pricing decisions.

9. Conclusion

This exploratory data analysis helps identify patterns and relationships between house prices and property features. The findings can support further analysis and the development of machine learning models for house price prediction.